When Post-Editing Helps and When It Harms: Evaluating LLM Post-Editing of NMT for Tagalog and Spanish

Anonymous working manuscript, 31 October 2026. Protocol frozen on development data on 24 October 2026; all results are from the held-out devtest set; paragraphs marked exploratory were added after the confirmatory analysis. The red [PENDING] note marks one open venue item.

***Abstract*—Large language models (LLMs) can post-edit neural machine translation (NMT) output, yet evaluations that report average gains can conceal damage to drafts that were already correct. This study asks when LLM post-editing helps and when it harms a strong NMT baseline for English–Tagalog and English–Spanish in both directions. A sentence-split NLLB-200 3.3B system supplies drafts for the held-out FLORES+ devtest set (1,012 sentences per direction); Llama 3 8B Instruct and GPT-4o Mini edit them zero-shot, with one retrieved development exemplar, or with a single-call reason-then-edit format. Prompts, output contracts, and analyses were frozen on development data before the test set was opened, and all 24,288 outputs are evaluated with paired, multiplicity-corrected tests and a preservation analysis of strong drafts. GPT-4o Mini improves COMET-22 in all twelve conditions (+0.26 to +1.00 points, all significant after Holm correction), most with the retrieved exemplar. Llama 3 8B never improves significantly; it is significantly worse in six conditions, and its reason-then-edit outputs degrade 247 of 1,179 strong drafts (20.9%), against at most 4.3% for any GPT-4o Mini strategy. Gains come from weak and mid-quality drafts, whereas strong drafts are at best preserved. Preserving correct drafts, not edit frequency or prompting strategy, separates a useful post-editor from a harmful one.**

***Index Terms*—automatic post-editing, machine translation, Tagalog, Spanish, over-editing, LLM evaluation**

I. Introduction

An automatic post-editor must decide both what to correct and what to leave alone. Large language models (LLMs) can repair neural machine translation (NMT) output [1]–[3], but a fluent rewrite can also alter a name, number, or source relation that the draft already expressed correctly, and LLM editors are known to produce hallucinated edits [1].

Problem. When drafts come from a strong multilingual NMT system, most sentences need little or no change. An editor can then raise the average score while damaging many correct drafts, and a mean difference cannot reveal this trade-off. For a lower-resource pair such as English–Tagalog, evidence is limited on whether an LLM editor adds value or risk on top of a strong baseline, and on whether the answer depends on the editor, the prompting strategy, or the translation direction. In deployment, an editor that silently degrades correct output is worse than none.

Approach. This study evaluates the trade-off for English–Tagalog and English–Spanish in both directions. A fixed NLLB-200 3.3B system [4] produces sentence-split drafts, and two instruction-following editors, open-weight Llama 3 8B Instruct and commercial GPT-4o Mini, receive the source and the draft. The two language pairs provide a limited resource contrast; language, direction, domain, and baseline quality remain intertwined, so pair and direction differences are reported descriptively rather than as causal resource effects. The study addresses four questions: **RQ1:** How do editing gains and harms differ between the Tagalog and Spanish pairs, conditional on baseline quality? **RQ2:** How do the three prompting strategies compare within each model and direction? **RQ3:** When do editors change already strong drafts and degrade their measured quality or source fidelity? **RQ4:** How do effects differ between the two directions of a language pair?

This is an empirical evaluation, not a new post-editing architecture. Anchoring an LLM editor on a frozen NMT draft and selecting in-context exemplars by semantic similarity follow established prior work [5], [6], and we do not claim either as a contribution. Our contributions are:

- a held-out evaluation in a new empirical setting (four Tagalog/Spanish directions) against a stronger, sentence-split NLLB-200 baseline, with the protocol frozen on development data before the test partition was opened;
- a two-editor comparison (open-weight and commercial) across three prompting strategies, with paired confirmatory tests and Holm multiplicity control;
- explicit failure accounting: every planned output is retained, and output-contract failures are reported as outcomes rather than discarded;
- an over-editing and preservation analysis of strong drafts that separates metric-defined harm from heuristic preservation screens.

The central finding is that post-editing helps only when the editor leaves adequate drafts alone; neither edit frequency nor explicit reasoning explains the difference between the editors.

II. Related Work and Research Gap

LLM post-editing. Raunak et al. [1] formalize direct translation post-editing with GPT-4 and report both quality gains and hallucinated edits. Ki and Carpuat [2] guide LLM post-editing with external error annotations, and xTower [3] trains a model to explain and correct translation errors, with expert assessment of its explanations. The reason-then-edit

condition studied here differs: an unmodified editor writes its own brief diagnosis in the same call, and that text is treated as an observable output, not as a verified explanation. LangMark [7] provides human post-edit triplets; FLORES+ references are independent translations, so the reference-based gains reported here are quality proxies and do not measure reduced human editing effort.

Exemplar selection. Vilar et al. [6] analyse prompting and example selection for translation. This study retrieves one development exemplar with LaBSE [8]; the similarity is semantic, and one deterministic exemplar is a reproducibility choice, not a claim of optimality.

Relationship to the ICAI study. Ahmadnia et al. [5] study English-to-German test-time domain adaptation with a frozen OPUS-MT draft, GPT-4 Turbo, multilingual Sentence-BERT top-three exemplars, and an optional second terminology-verification call. Frozen-NMT anchoring and semantic exemplar retrieval are shared with that work and are cited here as established methodology. The present study changes the empirical setting (four Tagalog/Spanish directions, FLORES+), the baseline (sentence-split NLLB-200 3.3B), and the editors (one open-weight and one commercial), uses a single-call reason-then-edit instead of a second verification call, and adds explicit failure and over-editing analysis. Because data, models, and metrics differ, no numerical comparison is made across the two papers.

Evaluation. COMET-22 [9], [10] is the primary metric, with BLEU [11] and chrF++ [12], [13] reported with full signatures [14].

Research gap. These studies show that LLM edits can improve NMT output and can be unfaithful. Less documented is how often an editor degrades already strong drafts, measured under a protocol fixed before test data are seen, for a lower-resource pair, with open-weight and commercial editors on identical inputs; this study addresses that gap.

III. Study Design and Systems

Design. The study is a fully crossed, paired comparison: two editors $\times$ three strategies $\times$ four directions give 24 post-editing conditions, each applied to the same held-out S0 drafts and compared with S0 on identical sentence IDs.

NMT baseline (S0). Drafts come from facebook/nllb-200-3.3B (FP16; four beams, no sampling, early stopping, length penalty 1.0, maximum length 512; inputs checked rather than truncated). The primary baseline segments each source with pysbd 0.3.4 (character spans, `clean=False`; English rules for Tagalog as a documented heuristic, with title, clock, *atbp.*, and parenthetical safeguards), translates multi-sentence items fragment by fragment, and joins the fragments with one space; single-sentence items reuse the unsplit hypothesis. Segmentation is lossless by construction. The unsplit system is kept as a diagnostic, outside the confirmatory tests.

Editors. The editors are the original Meta-Llama-3-8B-Instruct checkpoint (BF16, vLLM 0.11.2, pinned revision) [15] and the dated gpt-4o-mini-2024-07-18 snapshot [16]. Both decode with temperature 0, top-p 1, seed 42, and a 1,024-token output cap. Every request carries the same system instruction: the source is authoritative; source-supported names, numbers, dates, and meaning must be preserved; stylistic rewriting is discouraged; and an accurate, natural, grammatical draft must be returned unchanged. Source and draft are passed as delimited data.

Strategies. *Zero-shot* returns one JSON field, `edited_translation`. *Retrieved one-shot* adds one solved development example (its source, NLLB draft, and reference) before the query. *Reason-then-edit* returns a free-text `diagnosis` followed by `edited_translation` in a single call; only the translation is scored. The current query's reference never enters any prompt.

Retrieval. LaBSE [8] embeds all 997 development sources per source language (normalized, 768 dimensions); an exact FAISS inner-product index returns the most similar development item, excluding exact duplicates of the query source and breaking ties by the smaller sentence ID.

Output contract and fallback. Structured decoding (vLLM; OpenAI strict JSON schema) constrains the response shape, and a strict validator then requires valid JSON without duplicate keys, the exact field set and order, non-empty fields, and a normal stop. Otherwise, the raw response is retained and the system output is the unchanged baseline draft:

$$\hat{y}_i = \begin{cases} e_i, & \text{if the response passes the contract,} \\ b_i, & \text{otherwise.} \end{cases} \tag{1}$$

Failures are neither repaired nor selectively regenerated, so every evaluated condition is a deployable policy that includes its own failure behaviour.

IV. Data, Protocol, and Analysis

Data. The study uses FLORES+ [4], [17] at a pinned revision: 997 `dev` and 1,012 `devtest` sentences per direction. The dataset's `fil_Latn` configuration maps to NLLB's `tgl_Latn`. Spanish is the benchmark's standard variety, and no claim is made about heritage, dialectal, or code-switched Spanish. All prompt development, retrieval, and pilots used `dev` only.

Development pilot and protocol revision (development evidence). On the same 100 development IDs in all four directions (2,400 outcomes), the measured first contract asked reason-then-edit for one to six unique error categories with short notes. Llama fell back to the baseline on 227/400 such outputs (GPT-4o Mini: 8/400), almost entirely because of the diagnostic validators (e.g., repeated categories), not because of JSON parsing (399/400 parsed). Because a condition that mostly returns the baseline is not a clean strategy comparison, the minimal contract described in Section III was adopted and fixed together with a reliability gate (at most 20/400 fallbacks per model and 10/100 per direction) before rerunning the affected development conditions. Under the revised contract, neither editor fell back on any of its 400 development outputs (Llama 0/400, GPT-4o Mini 0/400), so the gate passed.

Freeze. Prompts, schemas, model revisions, decoding, retrieval policy, output contract, statistical families, thresholds,

and code hashes were recorded in a freeze manifest before `devtest` was opened. Because the original split-generation code was not recovered, S0 was reconstructed from the recorded specification. The reconstruction reproduced all 2,991 saved development segmentations and the saved unsplit drafts exactly, whereas the reconstructed fragment generation reproduced 412/421 saved multi-sentence development translations; the residual is attributed to batch-dependent FP16 beam search, and the earlier development baseline was not regenerated. On `devtest`, single-sentence items reuse the checksum-locked unsplit drafts.

Metrics and tests. COMET-22 (Unbabel/wmt22-comet-da, pinned checkpoint) is primary and reported as native score $\times 100$. BLEU (13a, mixed case, exponential smoothing) and chrF++ (character order 6, word order 2, β=2) are descriptive. For a comparison, d_i is the native segment COMET difference on the same sentence ID; the analysis reports $100\bar{d}$, a paired percentile 95% interval from 10,000 resamples, and a two-sided centred paired-bootstrap p-value, $p = (1+\#\{|\bar{d}_0^*| \geq |\bar{d}|\})/(B+1)$. Holm correction is applied within two prespecified families of 24 tests: each condition against S0 (F1), and one-shot−zero-shot, reason−zero-shot, and reason−one-shot within each editor and direction (F2). Non-significant differences are inconclusive; an interval inside ± 0.5 points is flagged as practically small, not as equivalence. Pair and direction differences (RQ1, RQ4) are described from these estimates without separate tests.

Harm and over-editing. A segment improves if $d_i > 0.02$ (two reported points), is harmed if $d_i < -0.02$, and otherwise lies within the band. The protect pool contains items whose S0 COMET is at least 0.90 and the improve pool items below 0.80, both fixed on S0 before editing. A protect-pool over-edit is a textual change (NFC, outer whitespace removed) with $d_i < -0.02$, reported over all protect items, with sensitivity at 0.01 and 0.05. Digit, capitalized-token, language-identification (langdetect), length-ratio, and sentence-count screens are heuristics, not gold error labels.

V. Held-out Results

Results follow the research questions, after the overall comparison with S0. All 24 conditions contain the planned 1,012 items per direction, and every output passed the output contract: no condition used the baseline fallback (Table I). The measured effects are therefore editing behaviour, not contract failure. As a diagnostic outside the test families, sentence splitting itself raised S0 above the unsplit system by $+0.39$ (En→Tl), $+0.21$ (En→Es), and $+0.08$ points (Tl→En); Es→En was inconclusive.

Overall effect against S0 (F1). GPT-4o Mini improves on S0 in every direction and strategy, by +0.26 to +1.00 points, all significant after Holm correction; five of these intervals, all in the Spanish pair, lie within ± 0.5 points, so several gains are clear but small. Llama 3 8B never improves significantly. Its zero-shot and one-shot outputs are significantly worse only for En→Tl (-0.48 and -0.39), with the other six intervals inside ± 0.5 points, and reason-then-edit is significantly worse in all four directions (-0.51 to -1.44). BLEU and chrF++ lie above

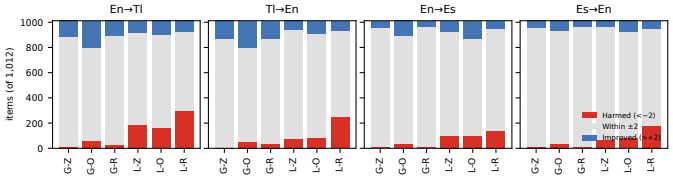


Fig. 1. Improved (>+2 points), within-band, and harmed (< −2) devtest items per condition (G/L: GPT-4o Mini/Llama; Z/O/R: zero-shot/one-shot/reason-then-edit). Unchanged outputs lie within the band.

S0 for all twelve GPT-4o Mini conditions and below it for all twelve Llama conditions, including those where COMET does not change significantly.

RQ1: pairs and baseline quality. GPT-4o Mini gains are larger for the Tagalog pair (+0.51 to +1.00) than for the Spanish pair (+0.26 to +0.53), and Llama losses are also largest for Tagalog. The Tagalog directions, however, contain more weak drafts (S0 COMET < 0.80: 183 and 82 items, against 104 and 60). Within this improve pool, GPT-4o Mini one-shot gains are similar across pairs (+3.11 to +5.29 points), while in the protect pool (S0 ≥ 0.90) every GPT-4o Mini condition stays near zero (-0.16 to $+0.15$). Baseline quality explains only part of the pair difference, however: reweighting each direction to the pooled weak/middle/strong mix narrows GPT-4o Mini's Tagalog–Spanish gap by 16–23%, and at least 73% of the remaining gap arises in mid-quality drafts (exploratory).

RQ2: strategies (F2). 15 of 24 strategy contrasts are significant after Holm correction. For GPT-4o Mini, the retrieved exemplar beats zero-shot in three directions (+0.17 to +0.31; Es→En inconclusive), and reason-then-edit falls below one-shot in the same three (-0.27 to -0.44) but below zero-shot only for Tl→En (-0.18). For Llama, one-shot and zero-shot do not differ significantly in any direction, whereas reason-then-edit is significantly worse than both in all four (-0.51 to -1.35).

RQ3: over-editing of strong drafts. Edit rate alone does not predict harm (Fig. 1, Table II). GPT-4o Mini one-shot changes 3,249 of 4,048 drafts (80%) yet over-edits only 51 of 1,179 strong drafts (4.3%); its zero-shot and reason-then-edit rates are 0.8% and 2.1%. Llama edits at similar rates but over-edits 11.2% (zero-shot), 11.3% (one-shot), and 20.9% (reason-then-edit), up to 107 of 377 strong Tl→En drafts. Llama harms more segments than it improves in 9 of 12 conditions; GPT-4o Mini improves more than it harms in all 12.

RQ4: direction. Direction effects are small and inconsistent: GPT-4o Mini gains more into English than out of English for every strategy in the Tagalog pair (e.g., $+0.83$ vs. $+0.64$ zero-shot), but not in the Spanish pair (one-shot $+0.36$ into English vs. $+0.53$ out of English).

Robustness (exploratory). No F1 decision changes under Bonferroni or Benjamini–Hochberg correction, or under Holm over both families jointly, and all 18 significant F1 effects have the same sign in COMET, BLEU, and chrF++. Wilson 95% intervals keep the pooled over-editing rates apart (GPT-4o Mini at most 5.6%, Llama at least 9.5%), and among edited drafts

TABLE I

HELD-OUT FLORES+ DEVTEST RESULTS (1,012 ITEMS PER DIRECTION; ALL PLANNED OUTPUTS, INCLUDING BASELINE FALLBACK). C: COMET-22×100; Δ: POST-EDIT MINUS SENTENCE-SPLIT NLLB (S0) WITH PAIRED 95% BOOTSTRAP INTERVAL; MARKERS USE HOLM-ADJUSTED p OVER THE 24-TEST PRIMARY FAMILY ($^{\dagger}p < .05$, $^{\ddagger}p < .01$). FB: BASELINE FALLBACKS. BLEU/CHRF++ ARE DESCRIPTIVE.

		En→Tl			Tl→En			En→Es			Es→En		
Editor	Strategy	C	Δ [95% CI]	FB	C	Δ [95% CI]	FB	C	Δ [95% CI]	FB	C	Δ [95% CI]	FB
S0 split NLLB		85.20	–	–	87.57	–	–	86.67	–	–	87.37	–	–
GPT-4o Mini	Zero	85.84	+0.64‡ [+0.50, +0.77]	0	88.39	+0.83‡ [+0.69, +0.97]	0	86.95	+0.28‡ [+0.17, +0.39]	0	87.71	+0.34‡ [+0.24, +0.46]	0
	One	86.15	+0.95‡ [+0.78, +1.12]	0	88.57	+1.00‡ [+0.83, +1.18]	0	87.20	+0.53‡ [+0.37, +0.69]	0	87.73	+0.36‡ [+0.24, +0.50]	0
	Reason	85.71	+0.51‡ [+0.38, +0.64]	0	88.22	+0.65‡ [+0.50, +0.81]	0	86.93	+0.26‡ [+0.14, +0.38]	0	87.64	+0.27‡ [+0.17, +0.39]	0
Llama 3 8B	Zero	84.72	-0.48‡ [-0.65, -0.32]	0	87.47	-0.10 [-0.26, +0.05]	0	86.68	+0.00 [-0.18, +0.19]	0	87.36	-0.01 [-0.13, +0.12]	0
	One	84.81	-0.39‡ [-0.57, -0.21]	0	87.60	+0.03 [-0.12, +0.18]	0	86.85	+0.18 [-0.05, +0.40]	0	87.33	-0.04 [-0.24, +0.16]	0
	Reason	83.76	-1.44‡ [-1.70, -1.21]	0	86.25	-1.32‡ [-1.57, -1.07]	0	86.16	-0.51‡ [-0.71, -0.31]	0	86.63	-0.74‡ [-0.93, -0.55]	0

TABLE II

PROTECT-POOL OVER-EDITING ON DEVTEST: OVER-EDITED ITEMS / PROTECT-POOL ITEMS PER DIRECTION (PROTECT: S0 COMET≥0.90; OVER-EDIT: TEXT CHANGED AND NATIVE COMET LOSS >0.02).

Editor	Strategy	En→Tl	Tl→En	En→Es	Es→En
GPT-4o Mini	Zero	3/234	3/377	3/275	1/293
	One	6/234	26/377	7/275	12/293
	Reason	5/234	13/377	3/275	4/293
Llama 3 8B	Zero	42/234	35/377	29/275	26/293
	One	35/234	37/377	28/275	33/293
	Reason	57/234	107/377	32/275	51/293

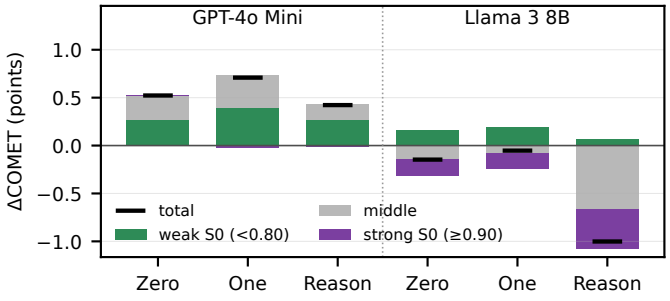


Fig. 2. Exact decomposition of the mean ΔCOMET into contributions of weak (S0 COMET < 0.80), middle, and strong (≥ 0.90) drafts, averaged over the four directions; the tick marks the total (exploratory).

every GPT-4o Mini condition harms at most 8.9% of them, below the 9.9% minimum for Llama.

Summary. GPT-4o Mini improved every condition and rarely damaged strong drafts; Llama improved none. The retrieved exemplar helped GPT-4o Mini most, and reason-then-edit ranked last for both editors in every direction.

VI. DISCUSSION

Preservation, not edit frequency, separates the editors (RQ3). In the protect pool, Llama's mean change ranges from -0.33 to -2.09 points across conditions, while its improve-pool changes are positive in 11 of 12 conditions: it repairs weak drafts but also rewrites good ones.

Where the gains come from (RQ1). An exact decomposition of each mean change by pool locates gains and losses (Fig. 2). Weak drafts make up only 6–18% of items per direction yet supply 41–77% of each GPT-4o Mini gain, while strong drafts contribute almost nothing. For Llama, losses on mid-quality and strong drafts outweigh its weak-draft repairs. Even an oracle gate that left strong drafts untouched (it needs the reference, so it is not deployable) would keep Llama below S0 in 6 of 12 conditions, because it also degrades mid-quality drafts. Protecting strong drafts is therefore necessary but not sufficient.

Explicit reasoning did not help (RQ2). Reason-then-edit was never significantly better than zero-shot or one-shot, its point estimate was lower in all 16 such contrasts, and for Llama it was the most harmful strategy in every direction. Every held-out reason-then-edit output passed the output contract, so these losses reflect editing rather than format failure. Diagnoses are model outputs, not verified explanations.

Source- and reference-based inspection. The inspection covered 96 held-out cases selected by fixed rules before any text was read (per condition: the largest strong-draft over-edit, the largest gain, and the first digit-loss and language-ID flags). All 12 of the largest Llama over-edits were genuine errors, for example rendering a plane's *secuestro* as "kidnapping" rather than "hijacking" under all three strategies, or turning "least interesting" into "not unwilling". In contrast, 10 of the 12 largest GPT-4o Mini over-edits were acceptable or even source-supported changes that COMET penalized, such as restoring "luxury" where S0 had "glory" (-9.1 points). The largest gains repaired clear S0 errors such as repetition loops, some only partly. Llama reason-then-edit sometimes returned the untranslated source while its diagnosis claimed a one-word fix; language identification places 20 of its 4,048 outputs in the source language. Most digit and language flags reflected formatting or names. This is a qualitative inspection, not a human evaluation; the cases illustrate mechanisms rather than error rates.

Implications. First, preservation of strong drafts should be reported alongside mean quality, since the editors differ far more in over-editing than in edit rate. Second, a small editor such as Llama 3 8B should edit only drafts that a reference-free quality estimator, tuned on development data, flags as weak; the oracle shows headroom, not an achievable gate. Third, one

retrieved exemplar is an inexpensive gain for a strong editor (about US$0.10 per 1,000 sentences for GPT-4o Mini at list rates), whereas a diagnosis step adds tokens without benefit. The Llama over-edits match the unfaithful edits reported for LLM post-editing [1], and GPT-4o Mini's penalized rewordings show why reference-based scores need source inspection.

VII. Limitations and Ethical Considerations

Two language pairs cannot identify a causal effect of resource level: language, direction, domain, and baseline quality are confounded. FLORES+ sources are largely English-original, so reverse directions involve translated source text. References are independent translations rather than minimal post-edits, so neither editing effort nor productivity is measured, and no bilingual human evaluation was conducted; COMET and the preservation screens are imperfect proxies. One open-weight checkpoint and one dated commercial snapshot do not represent all LLMs, pretraining exposure to the public benchmark cannot be ruled out, and deterministic settings do not guarantee identical remote behaviour. Fragment-level translation can lose cross-sentence context, and the reconstructed S0 fragment generation reproduced 412/421 multi-sentence development translations rather than all of them; every editor post-edits the same frozen devtest drafts, so the comparisons are unaffected.

The study involves no human participants; it uses public benchmark text and automatic evaluation only. Results do not establish suitability for medical, legal, or other consequential translation, and standard-Spanish results should not be generalized to community or heritage varieties. Code, prompts, frozen configuration, and numeric results are available (link withheld for anonymous review); benchmark text and generations are not redistributed. All GPT-4o Mini runs (development and held-out) cost about US$1.26 at list token rates; Llama ran on one A100. **[PENDING: AI-use disclosure, only if the ICNLP submission instructions require it]**

VIII. Conclusions and Future Work

On held-out FLORES+ data, LLM post-editing of a strong, sentence-split NLLB-200 baseline helped or harmed depending above all on the editor. GPT-4o Mini improved every direction, most with one retrieved exemplar; Llama 3 8B never improved and over-edited strong drafts, especially with reason-then-edit. Gains came from weak and mid-quality drafts rather than strong ones, the pair difference was only partly explained by baseline quality, direction effects were small, and explicit diagnosis improved neither editor. The study contributes a frozen, fully accounted four-direction evaluation and shows that average gains, preservation of strong drafts, and output validity are separate properties; all three should be reported before an LLM post-editor is deployed. Future work should test reference-free quality-estimation gates selected on development data, more language pairs and editors, document-level context, and bilingual human assessment.

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